Cloaking experiments with Nassar-type metamaterials
(notes)

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1 Simple homogenization schema of Nassar-type unit cells

1.1 Transformation elasticity and the BGM gauge

An elastic cloak is an annular coating $\{r_i < \|\mathbf{x}\| < r_c\}$ that renders a void $\{\|\mathbf{x}\| < r_i\}$ invisible to elastic waves. The constitutive properties of the coating follow from a blow-up coordinate transformation $\mathbf{x} = \boldsymbol{\phi}(\mathbf{X})$ that maps the origin to the circle of radius r_i [BGM09].

Under the Brun–Guenneau–Movchan (BGM) gauge, the transformed elasticity tensor and mass density are [NCH18]

$$c_{ijkl} = \frac{F_{jm}\, F_{ln}\, C_{imkn}}{J}\,, \qquad \rho = \frac{R}{J}\,, \tag{1}$$

where $\mathbf{F} = \partial\mathbf{x}/\partial\mathbf{X}$ is the deformation gradient, $J = \det \mathbf{F}$, $\mathbf{C}$ and R are the background stiffness and density. The transformed tensor $\mathbf{c}$ breaks the minor symmetries ($c_{ijkl} \neq c_{jikl}$) while preserving major symmetry ($c_{ijkl} = c_{klij}$). A material realising such a tensor is necessarily *polar* (it resists rotations via body torques) and *degenerate* (it admits a stressless collapse mechanism) [NCH18].

For the circular blow-up with an isotropic background (λ, μ), the effective stiffness in the local polar basis $(\mathbf{m}, \mathbf{n})$ = (radial, tangential) reads

$$\mathbf{c}_{\text{eff}} = \begin{bmatrix} (2\mu{+}\lambda)\, f & \lambda & 0 & 0 \\ \lambda & (2\mu{+}\lambda)/f & 0 & 0 \\ 0 & 0 & \mu/f & \mu \\ 0 & 0 & \mu & \mu f \end{bmatrix}, \qquad f = \frac{\|\mathbf{x}\| - r_i}{\|\mathbf{x}\|}\,, \tag{2}$$

in augmented Voigt ordering $[\sigma_{11}, \sigma_{22}, \sigma_{12}, \sigma_{21}]$. The effective density is position-dependent:

$$\rho_{\text{eff}} = \frac{r_c^2}{(r_c - r_i)^2}\, f\, R\,. \tag{3}$$

Near the inner boundary ($r \to r_i$, $f \to 0$) the radial stiffness and density vanish while the tangential stiffness diverges — perfect cloaking requires this singular behaviour, which can only be approximated by a finite lattice.

1.2 Homogenized cell model

Instead of modelling the discrete lattice, we discretise the cloak annulus into a regular Cartesian grid of $n_x \times n_y$ square cells (Fig. 1). Each cell whose centre lies in $\{r_i < r < r_c\}$ is assigned a piecewise-constant effective stiffness $\mathbf{c}_{\text{lat}}$. The density is set to (3) computed from the cell center coordinates.

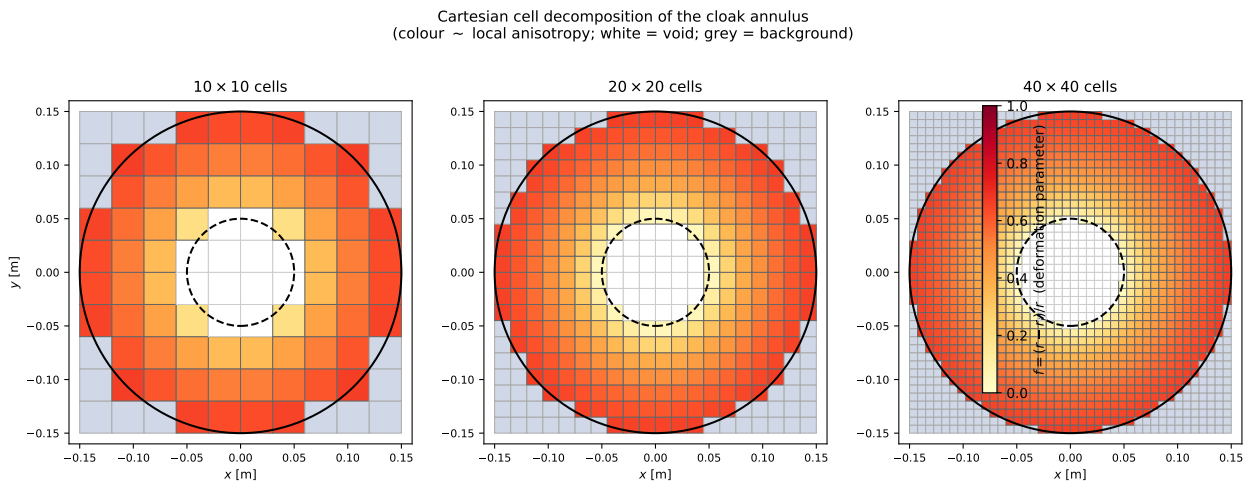

Figure 1: Cartesian cell decomposition of the circular cloak at different grid resolutions. Colour encodes the deformation parameter $f = (r - r_i)/r$, which controls the local anisotropy; $f \to 0$ near r_i (near-singular) and $f \to (r_c - r_i)/r_c$ near r_c (close to background). White cells lie inside the void; grey cells are background. Dashed/solid circles mark $r = r_i$ and $r = r_c$.

2 Circular cloaking of plane waves

2.1 Simulation setup

We consider a rectangular domain of side $4r_c = 0.6\,\text{m}$ with PML absorbing layers on all four boundaries. A circular void of radius $r_i = 50\,\text{mm}$ is centred in the domain, coated by an annular cloak of outer radius $r_c = 150\,\text{mm}$. The background medium is isotropic with $\lambda = \mu = 2.5\,\text{MPa}$ and $R = 2.7\,\text{kg/m}^2$ (plane stress).

The excitation is a Gaussian body-force line source at $x = 0.15\,W$ from the left physical boundary, producing approximately cylindrical pressure waves at $f = 40\,\text{kHz}$ (P-wave wavelength $\lambda_P \approx 42\,\text{mm}$, giving a void diameter of $\approx 2.4\,\lambda_P$). The frequency-domain elastodynamic equations are solved with complex-valued DOFs and Rayleigh-damping PML via a JAX-FEM finite-element solver on a triangular mesh with distance-based refinement near the cloak boundaries ($\sim$246,000 elements, $\sim$124,000 nodes).

2.2 Cloaking loss metric

We define the cloaking distortion as the relative L^2 displacement mismatch on the right physical boundary (the shadow side behind the scatterer):

$$\mathcal{D} = 100\,\frac{\|\mathbf{u}_{\text{cloak}} - \mathbf{u}_{\text{ref}}\|_{L^2(\partial\Omega_{\text{right}})}}{\|\mathbf{u}_{\text{ref}}\|_{L^2(\partial\Omega_{\text{right}})}}\ \ [\%]\,, \tag{4}$$

where $\mathbf{u}_{\text{ref}}$ is the displacement in the intact background (no void) and $\mathbf{u}_{\text{cloak}}$ is the solution with void and cloak. Perfect cloaking yields $\mathcal{D} = 0$.

2.3 Results

Three configurations are compared: (i) uncoated void (void present, no cloak), (ii) Nassar cell model at varying resolutions, and (iii) continuous $\mathbf{c}_{\text{eff}}$ from (2) (the theoretical optimum for the FEM discretisation).

Table 1: Cloaking distortion $\mathcal{D}$ as a function of cell grid resolution.

Configuration	Cloak cells	$\mathcal{D}$ [%]
Uncoated void	—	70.31
Nassar 5×5	20	119.64
Nassar 10×10	68	127.11
Nassar 20×20	284	91.60
Nassar 40×40	1 124	73.74
Nassar 60×60	2 512	49.42
Nassar 80×80	4 468	45.88
Nassar 100×100	6 988	37.02
Nassar 150×150	15 716	33.77
Nassar 200×200	27 948	28.63
Continuous $\mathbf{c}_{\text{eff}}$	∞	27.51

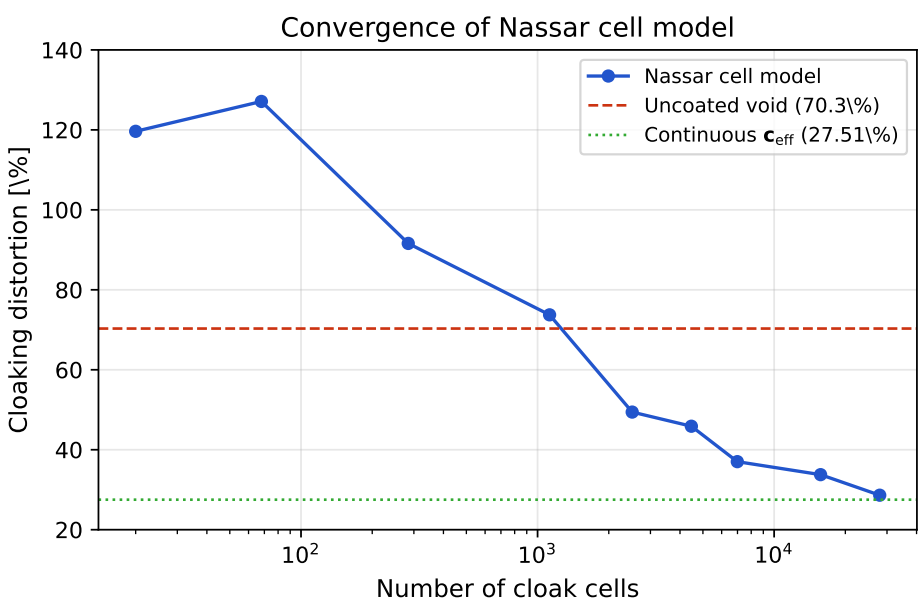

Figure 2: Convergence of the Nassar cell model (right-boundary loss). As the number of cloak cells increases, the distortion decreases from the coarse-grid value ($\mathcal{D} \approx 127\%$ at 10×10) toward the continuous limit ($\mathcal{D} \approx 27.5\%$). Above $\sim 60 \times 60$ cells the cloak outperforms the uncoated void (dashed red line), and at 200×200 cells the cell model approaches the continuous $\mathbf{c}_{\text{eff}}$ limit.

The residual $\sim 28\%$ distortion on the right boundary even for the continuous cloak is questionable and requires further investigation.

2.4 Mesh convergence of the continuous $\mathbf{c}_{\text{eff}}$ model

To understand the role of FEM discretisation error, the continuous $\mathbf{c}_{\text{eff}}$ cloak is solved on meshes ranging from 80×80 to 250×250 physical elements. Refinement x5 is applied near the circle. (This is not the cell

mesh from the experiment above, just the FEM mesh)

Table 2: Mesh convergence of the continuous $\mathbf{c}_{\text{eff}}$ model.

Mesh	$\mathcal{D}_{\text{right}}$ [%]	$\mathcal{D}_{\text{boundary}}$ [%]	$\mathcal{D}_{\text{outside}}$ [%]
80×80	41.58	4.32	4.22
120×120	31.70	2.49	2.35
160×160	27.51	1.33	1.57
200×200	27.13	1.27	1.30
250×250	23.02	1.04	1.06

A power-law fit $\mathcal{D}(h) = \mathcal{D}_\infty + A\,h^p$ to the right-boundary distortion yields convergence exponent $p \approx 1.35$ and asymptote $\mathcal{D}_\infty \approx 19\%$ (Fig. 3). This finding indicates that a significant portion of the $\sim 28\%$ residual distortion observed at 160×160 is FEM discretisation error rather than an intrinsic limitation of the continuous cloak (probably related to the discontinuity near the circle).

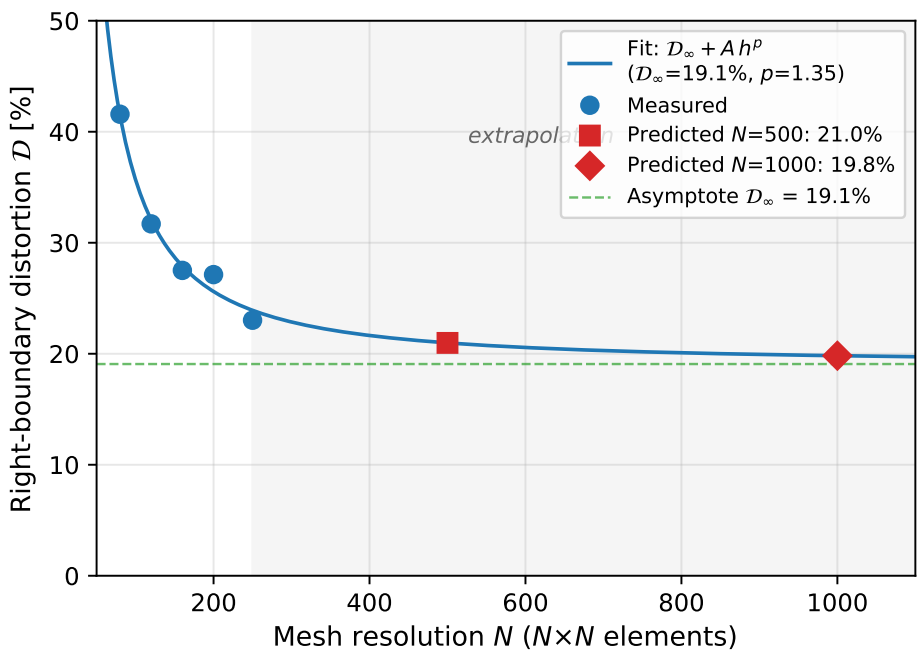

Figure 3: Mesh convergence of the continuous $\mathbf{c}_{\text{eff}}$ model (right-boundary distortion). Circles: measured data; solid line: power-law fit $\mathcal{D}_\infty + A\,h^{1.35}$ with $\mathcal{D}_\infty \approx 19\%$; diamond/square: extrapolated predictions. The grey region marks the extrapolation range beyond the finest computed mesh (250×250). Note that the fit has limited data points ($n = 5$) and the extrapolation carries substantial uncertainty ($\mathcal{D}_\infty = 19 \pm 5\%$).

2.5 Global outside-cloak distortion

In addition to the right-boundary metric $\mathcal{D}$, we evaluate the *outside-cloak distortion* $\mathcal{D}_{\text{out}}$, defined over all physical-domain nodes outside the cloak annulus:

$$\mathcal{D}_{\text{out}} = 100\,\frac{\|\mathbf{u}_{\text{cloak}} - \mathbf{u}_{\text{ref}}\|_{L^2(\Omega_{\text{phys}}\setminus\overline{\Omega}_{\text{cloak}})}}{\|\mathbf{u}_{\text{ref}}\|_{L^2(\Omega_{\text{phys}}\setminus\overline{\Omega}_{\text{cloak}})}}\ [\%]\,. \tag{5}$$

This metric captures the global wavefield perturbation due to the scatterer, rather than only the shadow-side mismatch. It is closely related to the objective function used by Wang et al. [Wan+22] in their data-driven

aperiodic metamaterial cloak design, where the L^2 displacement error over the exterior domain serves as the optimisation target for inverse-designed cloaks.

Table 2 shows that $\mathcal{D}_{\text{out}}$ converges more rapidly than the right-boundary metric and is already below 1.1% at 250×250. A power-law fit $\mathcal{D}_{\text{out}}(n) = \mathcal{D}_\infty + A\,n^{-p}$ (where n is the number of outside-cloak nodes) yields $p \approx 1.37$ and $\mathcal{D}_\infty \approx 0.75\%$ (Fig. 4). Extrapolation predicts $\mathcal{D}_{\text{out}} \approx 0.80\%$ at $N = 500$ and $\mathcal{D}_{\text{out}} \approx 0.76\%$ at $N = 1000$, confirming that the continuous $\mathbf{c}_{\text{eff}}$ cloak achieves near-ideal global cloaking (sub-1% outside distortion) once the mesh is sufficiently fine.

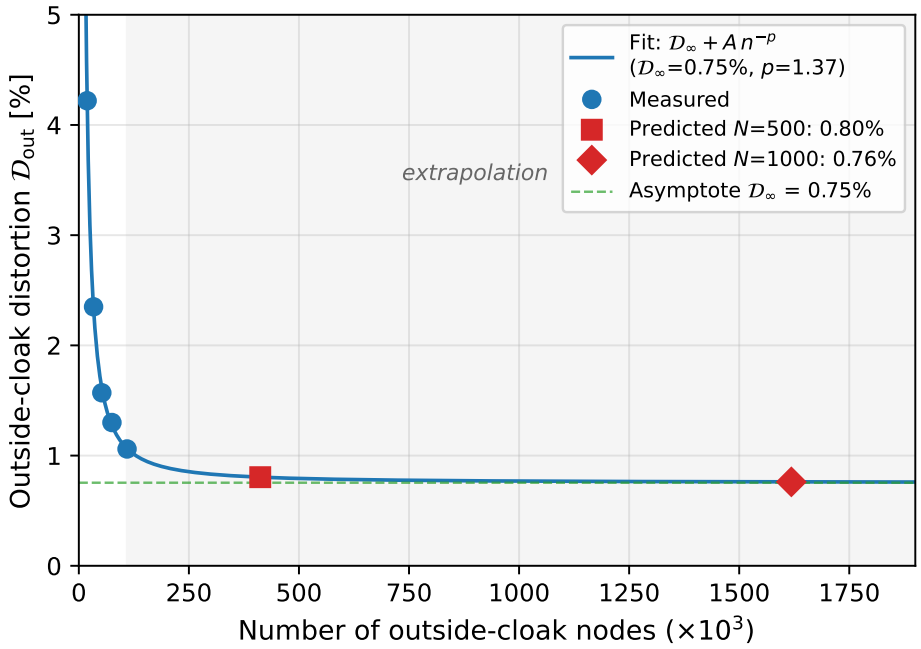

Figure 4: Mesh convergence of the outside-cloak distortion $\mathcal{D}_{\text{out}}$ for the continuous $\mathbf{c}_{\text{eff}}$ model, plotted against the number of outside-cloak nodes. Circles: measured data; solid line: power-law fit with $\mathcal{D}_\infty \approx 0.75\%$ and $p \approx 1.37$; diamond/square: extrapolated predictions. The asymptotic value $\mathcal{D}_\infty \approx 0.75\%$ serves as a reference target for the continuous cloak under this metric, comparable to the objective used in [Wan+22].

For comparison, Wang et al. [Wan+22] use the same type of metric — relative $L2$ displacement difference over the surrounding region Ω_s outside the cloak — in an elasto-static setting with a similar circular-void annular-cloak geometry ($r_c = 2\,r_i$). Their approach differs in two key respects: (i) they consider quasi-static loading (compression, pressure, shear) rather than frequency-domain wave propagation, and (ii) they work with classical Cauchy elasticity (orthotropic unit cells with four independent stiffness parameters $C_{11}, C_{12}, C_{22}, C_{33}$) rather than the Cosserat gauge with broken minor symmetry. Instead of analytical transformation-elasticity formulas, they employ a data-driven inverse design that selects aperiodic unit cells from a precomputed database. They report:

> *"With the designed cloak, the relative difference calculated from the numerical simulations is reduced from 17.2% in the voided structure to 4% in the cloaked structure, demonstrating a good cloaking performance."* — Wang et al. [Wan+22]

Under the analogous global metric $\mathcal{D}_{\text{out}}$, our continuous $\mathbf{c}_{\text{eff}}$ cloak reduces the outside distortion from 5.18% to an asymptotic $\sim 0.75\%$, substantially below the 4% achieved by Wang et al., though the comparison is not strictly apples-to-apples given the different physics.

3 Gradient-based optimisation of cell parameters

The entire pipeline — from cell material parameters through FEM assembly and solve to the cloaking loss — is implemented in JAX and is end-to-end differentiable via implicit adjoint differentiation through the linear FEM solve. This enables gradient-based optimisation of the per-cell stiffness and density to minimise the cloaking distortion directly.

We consider two parameterisation strategies for the material field over the 80×80 cell grid: (i) direct optimisation of raw per-cell parameters, and (ii) neural reparameterisation via a coordinate-conditioned MLP.

3.1 Loss function

The optimisation minimises a composite objective

$$\mathcal{L}(\mathbf{p}) = \mathcal{L}_{\text{cloak}}(\mathbf{p}) + \lambda_{\ell_2}\, \mathcal{L}_{\ell_2}(\mathbf{p}) + \lambda_{\text{nb}}\, \mathcal{L}_{\text{nb}}(\mathbf{p})\,, \tag{6}$$

where $\mathbf{p} = \{(\mathbf{C}_{\text{flat}}^{(k)}, \rho^{(k)})\}_{k=1}^{N_{\text{cells}}}$ collects all per-cell material parameters ($N_{\text{cells}} = 4\,468$ active cloak cells in the 80×80 grid, each with 10 stiffness + 1 density parameters $\Rightarrow \sim 49\,000$ optimisable DOFs).

The **cloaking term** measures relative L^2 displacement mismatch on the right physical boundary:

$$\mathcal{L}_{\text{cloak}} = \frac{\|\mathbf{u}_{\text{cloak}} - \mathbf{u}_{\text{ref}}\|^2}{\|\mathbf{u}_{\text{ref}}\|^2}\,, \tag{7}$$

where $\mathbf{u}_{\text{ref}}$ is the reference solution on the full (void-free) mesh and both solutions share the same mesh nodes on the measurement boundary (no interpolation needed).

The ℓ_2 **regularisation** penalises drift from the initial (transformation-theory) parameters:

$$\mathcal{L}_{\ell_2} = \frac{\|\mathbf{C}_{\text{flat}} - \mathbf{C}_{\text{flat}}^{(0)}\|^2}{\|\mathbf{C}_{\text{flat}}^{(0)}\|^2} + \frac{\|\boldsymbol{\rho} - \boldsymbol{\rho}^{(0)}\|^2}{\|\boldsymbol{\rho}^{(0)}\|^2}\,, \tag{8}$$

where each component is normalised by its own initial norm so that stiffness and density contribute equally.

The **neighbour smoothness** term penalises differences between 4-connected adjacent cloak cells:

$$\mathcal{L}_{\text{nb}} = \frac{1}{|\mathcal{N}|} \sum_{(i,j)\in\mathcal{N}} \left(\frac{\|\mathbf{C}_i - \mathbf{C}_j\|^2}{\langle \|\mathbf{C}\|^2 \rangle} + \frac{(\rho_i - \rho_j)^2}{\langle \rho^2 \rangle} \right), \tag{9}$$

where $\mathcal{N}$ is the set of neighbour pairs and the denominators are mean squared magnitudes for scale invariance.

Gradients $\partial\mathcal{L}/\partial\mathbf{p}$ are computed via JAX-FEM's `ad_wrapper`, which implements implicit adjoint differentiation through the sparse linear solve.

3.2 Approach 1: Direct per-cell optimisation

The most straightforward approach optimises the raw parameter arrays $\mathbf{C}_{\text{flat}}^{(k)} \in \mathbb{R}^{10}$ and $\rho^{(k)} \in \mathbb{R}$ for each cloak cell directly via Adam [KB15]. The transformational push-forward (2)–(3) evaluated at each cell centre provides the initialisation.

This representation has two drawbacks:

- **Memory.** The number of optimisable parameters scales linearly with the number of cells (and cubically with spatial resolution if the cell size is reduced to maintain accuracy). This is acceptable for small scatterers but does not scale to larger structures or 3D.
- **Smoothness.** Raw per-cell parameters do not enforce spatial continuity; the material field can develop cell-to-cell oscillations that are unphysical and degrade convergence. This is mitigated by the neighbour regularisation $\mathcal{L}_{\text{nb}}$ (with weight λ_{nb}), which acts as a discrete Laplacian penalty on the material field.

Results. Starting from the 80×80 Nassar cell initialisation (Table 1: $\mathcal{D}_{\text{right}} = 45.88\%$), Adam optimisation with $\text{lr} = 0.025$, $\lambda_{\ell_2} = 10^{-3}$, $\lambda_{\text{nb}} = 10^{-3}$ over 1 000 iterations produces:

Table 3: Cloaking distortion: analytical initialisation vs. optimised 80×80 cells vs. continuous $\mathbf{c}_{\text{eff}}$.

Configuration	$\mathcal{D}_{\text{right}}$ [%]	$\mathcal{D}_{\text{boundary}}$ [%]	$\mathcal{D}_{\text{outside}}$ [%]
Uncoated void	70.31	—	5.18
Nassar 80×80 (init)	45.88	—	—
Continuous $\mathbf{c}_{\text{eff}}$	27.51	1.33	1.57
Optimised 80×80 (direct)	**0.09**	**1.58**	**1.74**

The right-boundary distortion drops from 45.88% to 0.09% — essentially perfect cloaking on the shadow side — which is substantially better than the continuous $\mathbf{c}_{\text{eff}}$ model (27.51%). The boundary and outside-cloak metrics (1.58% and 1.74%, respectively) are comparable to the continuous model at the same 160×160 FEM mesh resolution (Table 2). This indicates that the optimiser exploits the additional degrees of freedom (broken minor symmetry, non-analytical density profile) to compensate for both the piecewise-constant discretisation error and the intrinsic limitations of the radial blow-up transformation.

3.3 Approach 2: Neural reparameterisation

Instead of optimising raw per-cell arrays, we reparameterise the material field through a small coordinate-conditioned MLP:

$$\Phi_\theta \colon (x, y) \;\mapsto\; (\mathbf{C}_{\text{flat}},\, \rho)\,, \tag{10}$$

where θ denotes the network weights. The MLP is evaluated at each cell centre to produce the material parameters, which are then expanded to FEM quadrature points exactly as in the direct approach. Optimisation proceeds over θ rather than the cell arrays; gradients flow through the FEM adjoint and then through the network via standard backpropagation.

This is conceptually analogous to the neural implicit representations used in computer graphics (NeRF [Mil+21] and related work) but applied to physical material fields rather than radiance. The key advantages are:

- **Implicit smoothness.** Neural networks are smooth functions of their inputs by construction, so the material field is spatially continuous without explicit neighbour regularisation ($\lambda_{\text{nb}} = 0$).
- **Compact representation.** A 4-layer MLP with 256 hidden units has $\sim$200k weights, which parameterises the $\sim$49k cell values through a lower-dimensional manifold that encodes spatial correlations.
- **Resolution-independent.** The same network can be evaluated at arbitrary cell centres, enabling coarse-to-fine transfer without retraining.

Following Sitzmann et al. [Sit+20], we use Fourier input features (sinusoidal positional encoding of the normalised cell coordinates) to allow the network to represent high-frequency spatial variations. SIREN-style periodic activations (sin instead of tanh) are a natural alternative that may further improve the representation of sharp material transitions near the cloak boundaries; this is left for future experiments.

Transformation prior. Crucially, the network does *not* learn the full material field from scratch. Instead, it learns a **residual correction** to the analytical transformation-elasticity solution. The last network layer is initialised near zero ($\mathbf{W} \leftarrow 0.01\,\mathbf{W}$, $\mathbf{b} \leftarrow \mathbf{0}$), so at step 0 the material field recovers the analytical initialisation exactly. Two scaling strategies for the residual are natural:

Additive (absolute) residual.

$$\mathbf{C}_{\text{flat}}(x, y) = \mathbf{C}^{(0)}_{\text{flat}}(x, y) + \epsilon\, \boldsymbol{\sigma}_C \odot \Phi^{(C)}_\theta(x, y)\,, \tag{11}$$

where $\boldsymbol{\sigma}_C$ is the per-component standard deviation of the initial field across cloak cells and ϵ is a global output scale. This is simple but problematic: the effective stiffness $\mathbf{c}_{\text{eff}}$ varies by orders of magnitude across the cloak (near-zero radial stiffness at $r \to r_i$, background-level at $r \to r_c$), so a single additive scale produces relative perturbations that are either too small where C is large or too large where C is small.

Multiplicative (relative) residual.

$$\mathbf{C}_{\text{flat}}(x,y) = \mathbf{C}^{(0)}_{\text{flat}}(x,y) \odot \left(1 + \epsilon\, \Phi^{(C)}_\theta(x,y)\right), \qquad \rho(x,y) = \rho^{(0)}(x,y)\left(1 + \epsilon\, \Phi^{(\rho)}_\theta(x,y)\right), \tag{12}$$

where ϵ is the output scale (default 0.1). Each cell's correction is proportional to its own initial value, so the relative perturbation is spatially uniform regardless of the local magnitude of $\mathbf{c}_{\text{eff}}$. This eliminates the need for per-component normalisation and produces well-conditioned gradients across the entire cloak. We use this form in practice.

Preliminary results. Neural reparameterisation has been implemented and produces valid gradients through the full pipeline. Systematic comparison with the direct approach (convergence speed, final loss, generalisation across frequencies) is ongoing. Additional regularisation strategies for material physical plausibility (e.g. positive-definite stiffness) are also being explored.

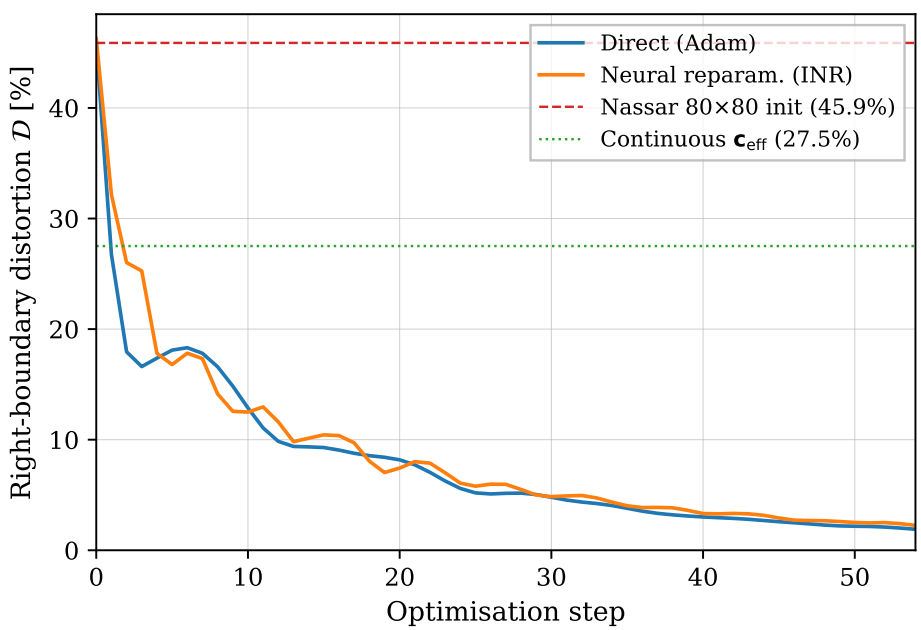

Figure 5: Convergence of right-boundary distortion $\mathcal{D}$ during optimisation for the direct (Adam) and neural reparameterisation (INR) approaches on the 80×80 grid. Both methods drop below the continuous $\mathbf{c}_{\text{eff}}$ baseline (27.5%) within ~ 10 steps and reach $\sim 3\%$ by step 50. Dashed/dotted lines mark the Nassar 80×80 initialisation and the continuous transformation-elasticity limit.

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